

MIAE 221 · Rapid Review Sheet · Part 5

Imperfections in Solids, Point Defects, Solid Solutions & Dislocations

1. Thermodynamics & Point Defects

- **Thermodynamic Law:** A defect-free crystal is impossible at $T > 0$ K. Creating vacancies raises enthalpy ($\Delta H > 0$), but introduces configurational entropy ($\Delta S > 0$), lowering Gibbs free energy ($G = H - TS$).
- **Vacancy:** Missing lattice site; produces localized inward tensile strain.
- **Self-Interstitial:** Host atom squeezed into small interstitial void; produces severe compressive strain ($Q_{\text{interstitial}} \gg Q_v \implies$ negligible concentration).

2. Equilibrium Vacancy Concentration (Arrhenius Relation)

$$\frac{N_v}{N} = \exp\left(-\frac{Q_v}{k_B T}\right)$$

- $N = \frac{\rho N_A}{A}$ = Total lattice sites/unit volume.
- Q_v = Vacancy formation energy (eV/atom or J/mol).
- $k_B = 8.62 \times 10^{-5}$ eV/atom · K = 1.38×10^{-23} J/atom · K.
- **Arrhenius Slope:** Plotting $\ln(N_v/N)$ vs. $1/T$ yields a straight line with Slope = $-Q_v/k_B$.

3. Hume-Rothery Rules for Substitutional Solid Solubility

For two metals to form extensive/complete substitutional solid solution:

1. **Atomic Size Factor:** $\Delta r = \frac{|r_{\text{solute}} - r_{\text{solvent}}|}{r_{\text{solvent}}} \times 100\% < 15\%$.
 2. **Crystal Structure Match:** Both metals must have the **same crystal structure** (e.g., both FCC for Cu and Ni).
 3. **Electronegativity Proximity:** $\Delta X = |X_{\text{solute}} - X_{\text{solvent}}| \leq \pm 0.4$. Large ΔX forms brittle intermetallic compounds.
 4. **Valency Rule:** Higher solubility when solute has equal or **higher valency** than solvent ($V_{\text{solute}} \geq V_{\text{solvent}}$).
- *Exemplary System:* Cu-Ni (FCC, $\Delta r = 2.3\%$, $\Delta X = 0.1 \implies 100\%$ miscible isomorphous system).

4. Composition Conversions

- $wt\% \rightarrow at\%: C'_1 = \frac{\frac{C_1}{A_1}}{\frac{C_1}{A_1} + \frac{C_2}{A_2}} \times 100\%$
- $at\% \rightarrow wt\%: C_1 = \frac{C'_1 A_1}{C'_1 A_1 + C'_2 A_2} \times 100\%$

5. Linear Defects: Dislocations & Plastic Slip

Dislocations allow atomic planes to slide one row at a time, lowering yield stress by $1000\times$:

Dislocation Type	Dislocation Line (t) vs. Burgers Vector (b)	Local Strain Field	Slip Motion vs. Shear Stress (τ)
Edge ($\top$)	$\mathbf{b} \perp \mathbf{t}$ (Perpendicular)	Compressive above slip plane, Tensile below	Parallel to τ
Screw	$\mathbf{b} \parallel \mathbf{t}$ (Parallel)	Pure Shear strain (helical spiral ramp)	Perpendicular to τ
Mixed	Angle $\theta \neq 0^\circ, 90^\circ$	Combined Compressive, Tensile & Shear	Intermediate angle

- **Invariance Law:** The Burgers vector **b** is **constant and uniform** along the entire length of any dislocation loop.